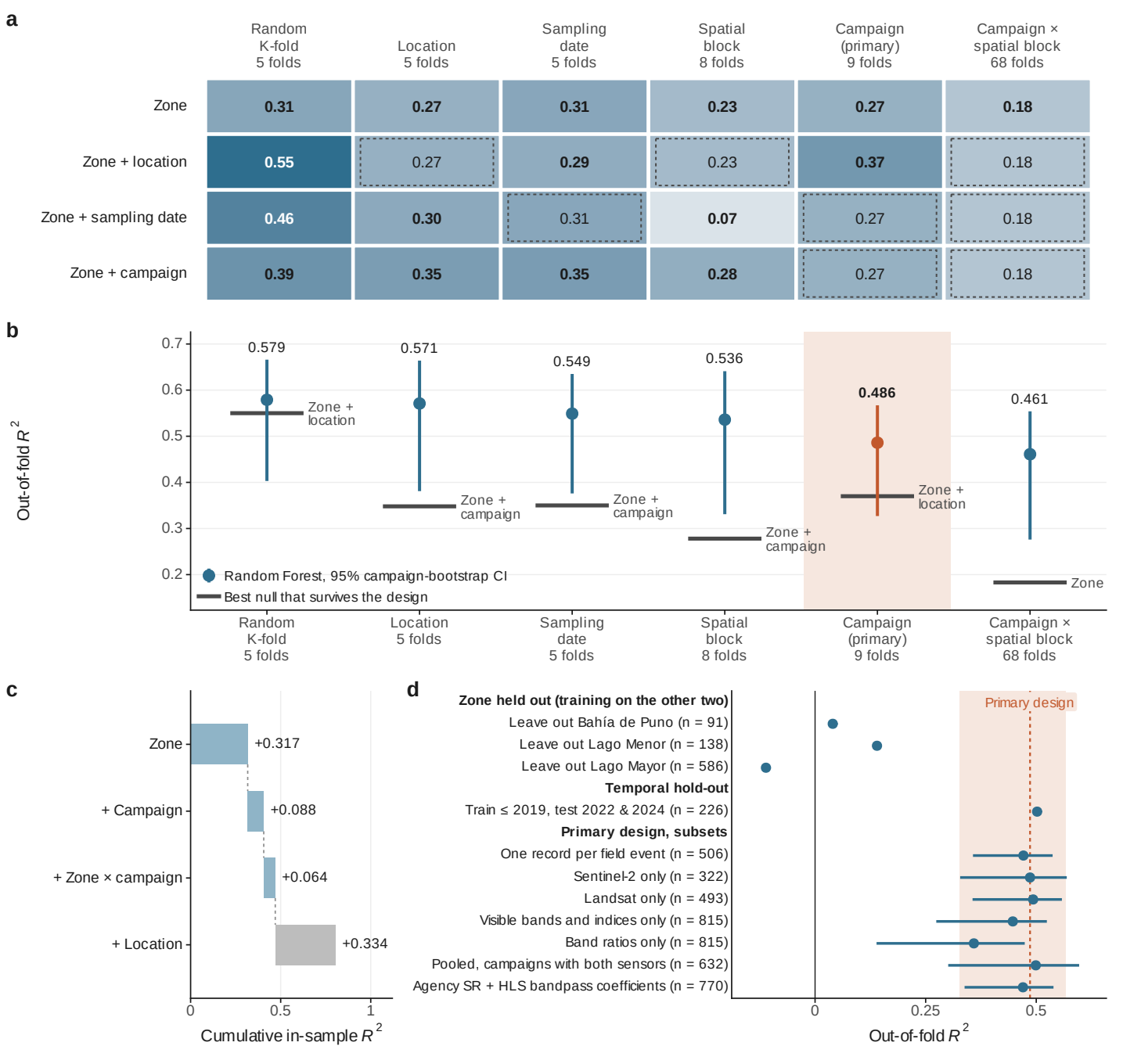
a
Random K-fold 5 folds
Location 5 folds
Sampling date 5 folds
Spatial block 8 folds
Campaign (primary) 9 folds
Campaign × spatial block 68 folds
Zone
Zone + location
Zone + sampling date
Zone + campaign
0.31
0.27
0.31
0.23
0.27
0.18
0.55
0.27
0.29
0.23
0.37
0.18
0.46
0.30
0.31
0.07
0.27
0.18
0.39
0.35
0.35
0.28
0.27
0.18
b
Out-of-fold R^2
0.579
0.571
0.549
0.536
0.486
0.461
Zone + location
Zone + campaign
Zone + campaign
Zone + campaign
Zone + location
Zone
Random Forest, 95% campaign-bootstrap CI
Best null that survives the design
c
Zone
+ Campaign
+ Zone × campaign
+ Location
+0.317
+0.088
+0.064
+0.334
Cumulative in-sample R^2
d
Zone held out (training on the other two)
Leave out Bahía de Puno (n = 91)
Leave out Lago Menor (n = 138)
Leave out Lago Mayor (n = 586)
Temporal hold-out
Train ≤ 2019, test 2022 & 2024 (n = 226)
Primary design, subsets
One record per field event (n = 506)
Sentinel-2 only (n = 322)
Landsat only (n = 493)
Visible bands and indices only (n = 815)
Band ratios only (n = 815)
Pooled, campaigns with both sensors (n = 632)
Agency SR + HLS bandpass coefficients (n = 770)
Primary design
Out-of-fold R^2